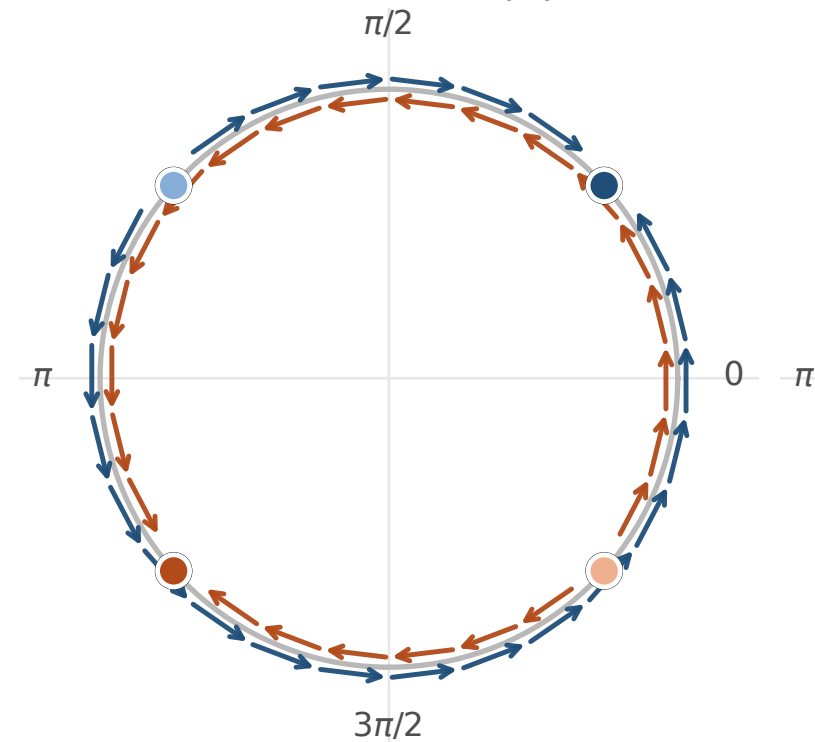
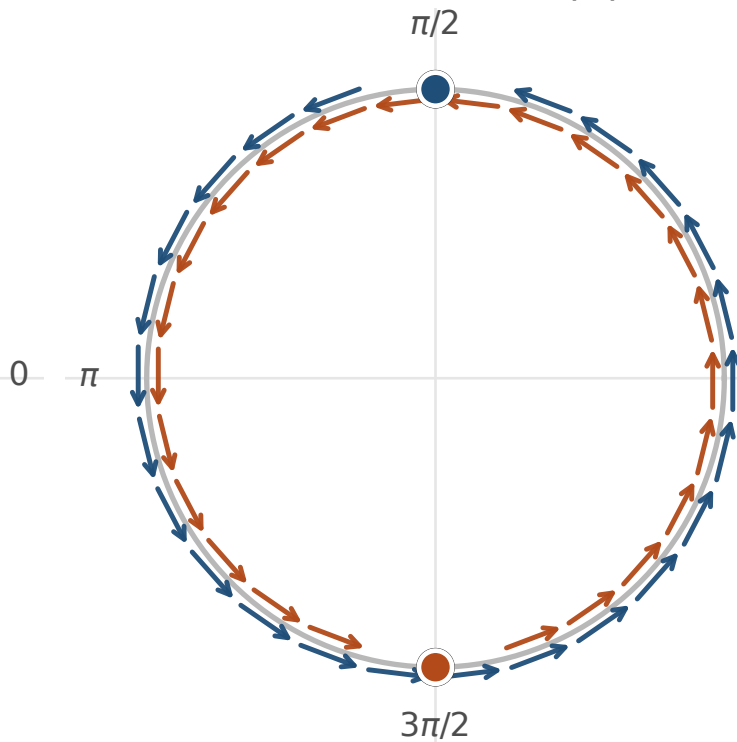


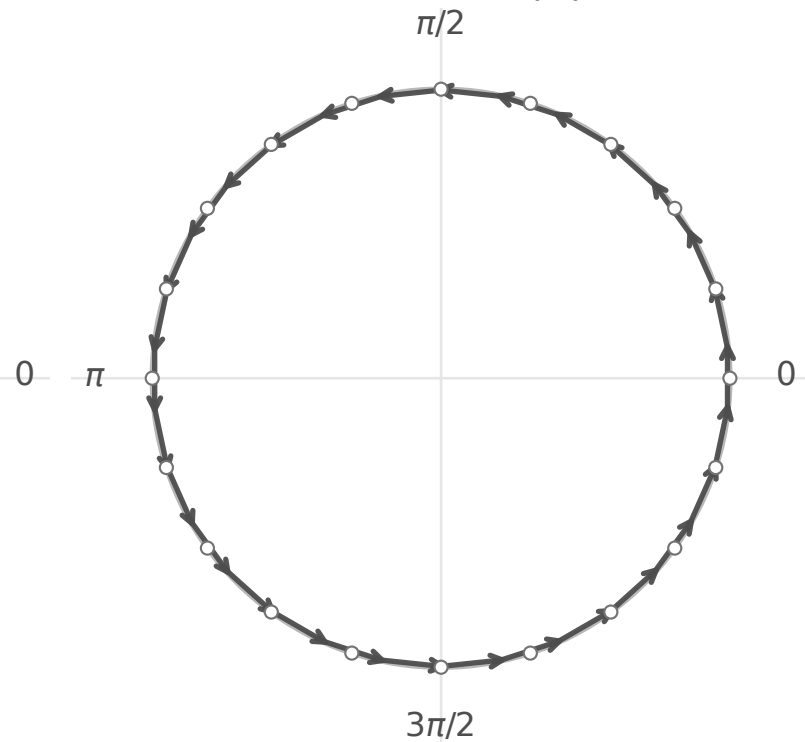
Two fixed points: $|a| > 1$



Critical fixed point: $|a| = 1$



No fixed point: $|a| < 1$



- $a > 1$: Stable FP, Q1
- $a > 1$: Unstable FP, Q2

- $a < -1$: Stable FP, Q3
- $a < -1$: Unstable FP, Q4

- $a = 1$: Critical FP at $\pi/2$
- $a = -1$: Critical FP at $3\pi/2$